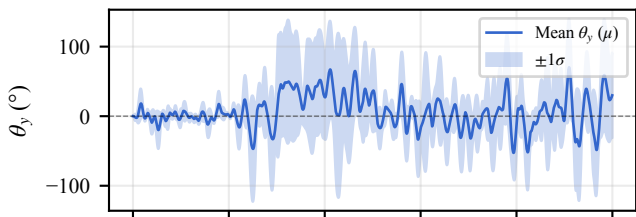
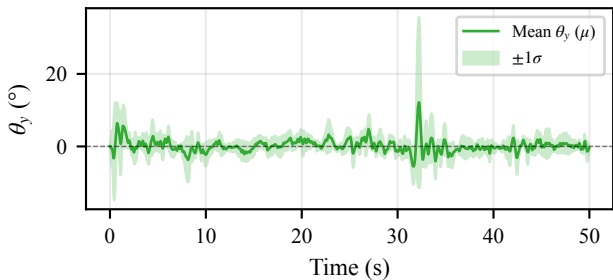


(a) Without attitude control (std=2103.3°, max=5497.7°)



(b) Cascaded PID (tuned at $r = 0.2$) (std=50.0°, max=203.5°)



(c) PID + residual RL (trained $r \leq 0.2$) (std=3.6°, max=52.7°)